A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

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INTRODUCTION

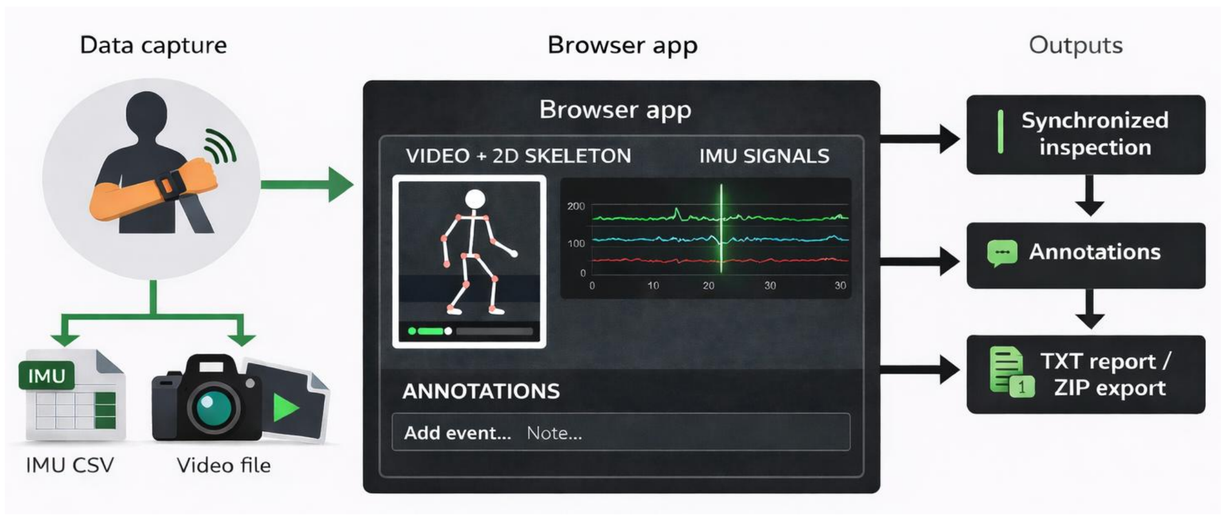
Motion tracking is widely used in healthcare, sports, and rehabilitation, but many high-end systems remain expensive and require controlled laboratory setups. A **browser-based workflow** can reduce cost and setup complexity while maintaining practical utility for inspection and analysis. The presented system combines **inertial measurement unit (IMU) signals** (accelerometer, gyroscope, magnetometer) with **video-based 2D pose estimation** and **event-based synchronization** to align both modalities in time.



OBJECTIVES

- **Objective:** Design a low-cost, low-complexity motion tracking prototype that runs in a standard web browser.
- **Aim:** Enable synchronized visual inspection of video-based pose estimation and IMU sensor data without specialized motion-capture hardware.

MATERIALS AND METHODS



SYSTEM OVERVIEW

Data are collected using an **IMU device** and a **standard video recording**. A **browser application** loads the video and IMU CSV data, performs pose estimation, visualizes signals, supports event annotation, and exports results for reproducible analysis.



IMU SUBSYSTEM

Movesense HR2 is used to capture **tri-axial accelerometer, gyroscope, and magnetometer signals**. Data logging was initially implemented in a Python workflow and later migrated to a browser-based pipeline **(HTML/JS with Chart.js**). **Timestamps** are **reconstructed** from the sample index using a fixed sampling rate of 104 Hz.

VIDEO SUBSYSTEM

Single-person 2D pose estimation is performed using **MoveNet Thunder** executed in **TensorFlow.js** for uploaded videos. To maintain stable performance on consumer hardware, the pipeline applies **frame skipping** (N = 2), **reduced playback speed** (v = 0.75×), and **keypoint filtering** (confidence c ≥ 0.2). These measures support responsive synchronized inspection during playback and scrubbing.

SYNCHRONIZATION

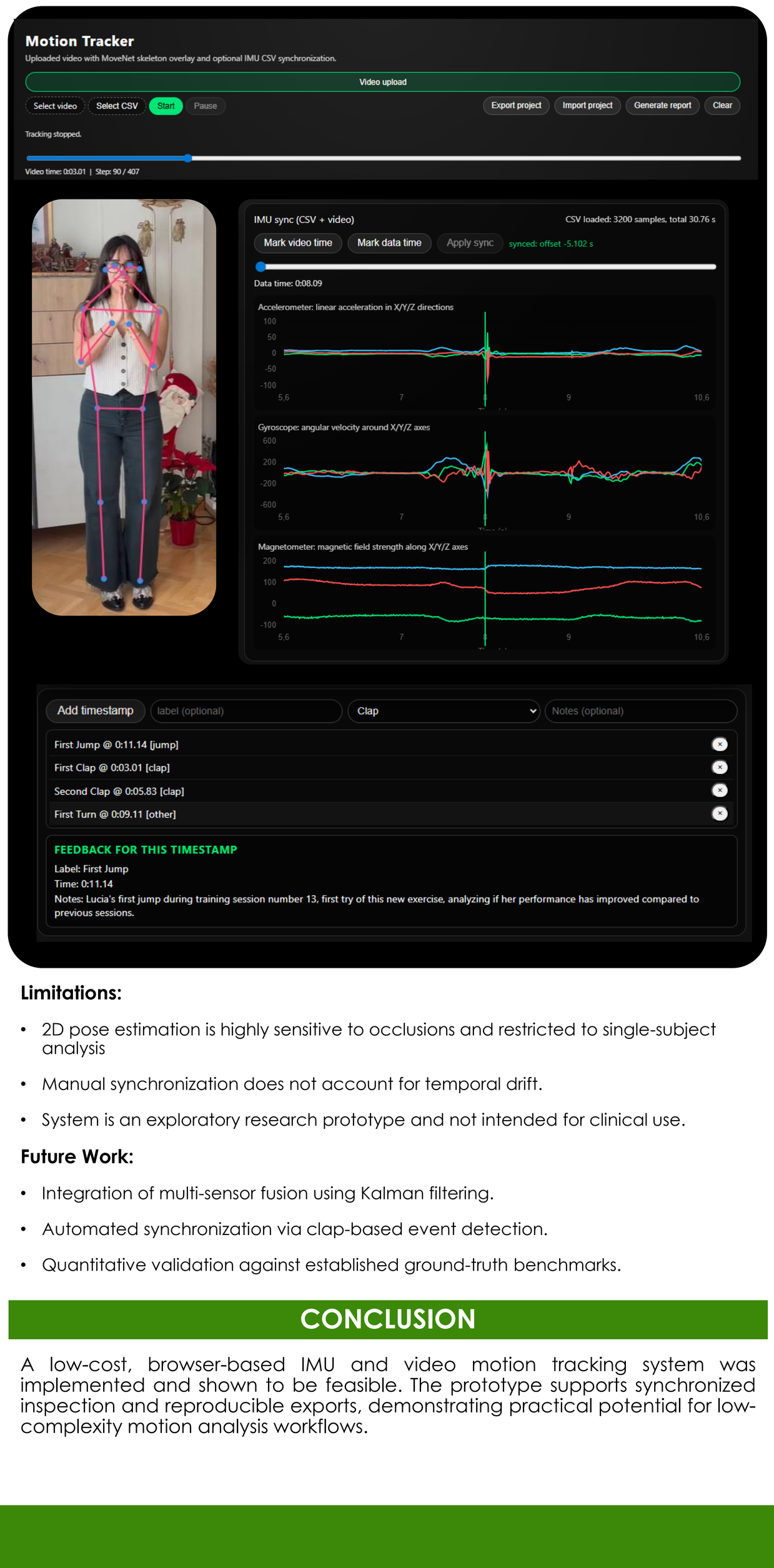
Temporal alignment between IMU signals and video is achieved by marking **matching reference events** across both modalities (e.g., claps or jumps). A constant offset is computed as $\Delta t = t_{\text{video}} - t_{\text{IMU}}$. The offset is applied to **synchronize timelines** during video playback and interactive scrubbing, enabling **coherent comparison** between the video skeleton overlay and IMU plots.

REPRODUCIBILITY

Synchronization offsets and annotations (time, label, notes) are **stored locally in the browser** (localStorage) to support repeatable workflows. ZIP export/import packages metadata together with video/CSV files to reproduce a **complete project state**. TXT reports summarize the IMU timeline alongside stored annotations.

RESULTS AND DISCUSSION

A functional browser-based IMU + video motion analysis prototype is implemented with **stable event synchronization** using the computed offset (Δt). The interface supports **synchronized playback, interactive scrubbing, and timestamped** annotations. Export functions produce **reproducible outputs** through ZIP packages and TXT summaries, supporting **repeated analysis** across sessions and **sharing between workflows**.



Limitations:

- 2D pose estimation is highly sensitive to occlusions and restricted to single-subject analysis
- Manual synchronization does not account for temporal drift.
- System is an exploratory research prototype and not intended for clinical use.

Future Work:

- Integration of multi-sensor fusion using Kalman filtering.
- Automated synchronization via clap-based event detection.
- Quantitative validation against established ground-truth benchmarks.

CONCLUSION

A low-cost, browser-based IMU and video motion tracking system was implemented and shown to be feasible. The prototype supports synchronized inspection and reproducible exports, demonstrating practical potential for low-complexity motion analysis workflows.

ACKNOWLEDGEMENTS

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